

A Leakage-Aware Benchmark for Ultra-Small Language Models on Structured Speech-Command Routing

[Removed for double-blind review]

¹[Removed for double-blind review]

Abstract. Deploying structured command routing locally on mobile devices requires ultra-small language models ($\leq 3B$ parameters). We present a reproducible benchmark evaluating five sub-3B instruction-tuned models across distinct families on a transcript-to-JSON routing task derived from Fluent Speech Commands (FSC). We evaluate under an explicit two-tool contract with balanced policy abstention (50% calls, 50% abstentions), comparing the official speaker-disjoint split against an audited phrase-disjoint partition ($n = 2,296$, 38 template clusters). Under the strict contract, a transparent lexical baseline achieves 77.40% exact match, outperforming every zero-shot neural model. Qwen3.5-2B attains 100% validity but collapses to universal abstention ($R_{call}^{exact} = 0\%$), while three models score 0% contract validity due to unconstrained schema deviations. We show this structural failure is resolved by supervised adaptation (LoRA), establishing crucial guidelines for edge SLM evaluation.

1. Introduction and Scope

Deploying natural language command routing locally on edge hardware requires ultra-small language models (SLMs, $\leq 3B$ parameters). However, evaluating these models zero-shot introduces two severe pitfalls: (i) template leakage across speakers in benchmark splits, and (ii) degenerate policy compliance, where models achieve seemingly high accuracy by trivial abstention.

We evaluate five checkpoints within the sub-3B on-device regime [4]: Qwen2.5-0.5B, Qwen3.5-0.8B/2B [7], TinyLlama-1.1B [5], and SmoLLM2-1.7B [6]. The input is a transcription from Fluent Speech Commands (FSC) [1], and the output is a validated JSON object conforming to a strict two-tool contract: `media_control` and `volume_adjust`, with all other 27 native intents mapped to a derived abstention target. The evaluation is balanced 1:1 (7,478 calls and 7,478 abstentions per split). We evaluate on both FSC’s official speaker-disjoint split ($n = 1,934$) and an audited phrase-disjoint split ($n = 2,296$, 38 template clusters) where no transcription template crosses partitions.

2. Experimental Results

As shown in Table 1 and Figure 1, the lexical control dominates all zero-shot models. Exact Call Recall (R_{call}^{exact}) denotes the ratio of correct calls with exact tool and arguments over gold call instances. Strict Template Exact denotes the proportion of template clusters where 100% of items are correctly routed. Qwen3.5-2B attains 100% validity and 50.00% exact match, but inspection reveals complete policy collapse: it outputs `abstain` on 100% of inputs, yielding $R_{call}^{exact} = 0\%$. Qwen3.5-0.8B is the only model attempting active

Table 1. Phrase-disjoint vs. Official test results (% , JSON = parseable JSON).

System	Phrase-Disjoint (Unseen Templates)				Official (Speaker Split)		
	JSON	Valid	Exact	R_{call}^{exact}	Valid	Exact	Template Exact
Always abstain	100.00	100.00	50.00	0.00	100.00	50.00	75.71
Lexical control	100.00	100.00	77.40	54.79	100.00	77.92	89.07
Qwen2.5-0.5B	70.17	0.00	0.00	0.00	0.00	0.00	0.00
Qwen3.5-0.8B	100.00	60.15	43.60	33.45	73.47	54.65	56.28
TinyLlama-1.1B	43.42	0.00	0.00	0.00	0.00	0.00	0.00
SmolLM2-1.7B	96.47	0.00	0.00	0.00	0.78	0.00	0.00
Qwen3.5-2B	100.00	100.00	50.00	0.00	100.00	50.98	76.11

routing, but exhibits an observed split gap of 11.05 pp from official (54.65%) to phrase-disjoint (43.60%). Because the phrase-disjoint cluster bootstrap 95% CI is wide ([26.02%, 63.35%], over 38 clusters with 9 call and 29 abstain clusters), this gap is descriptive rather than a formal causal test.

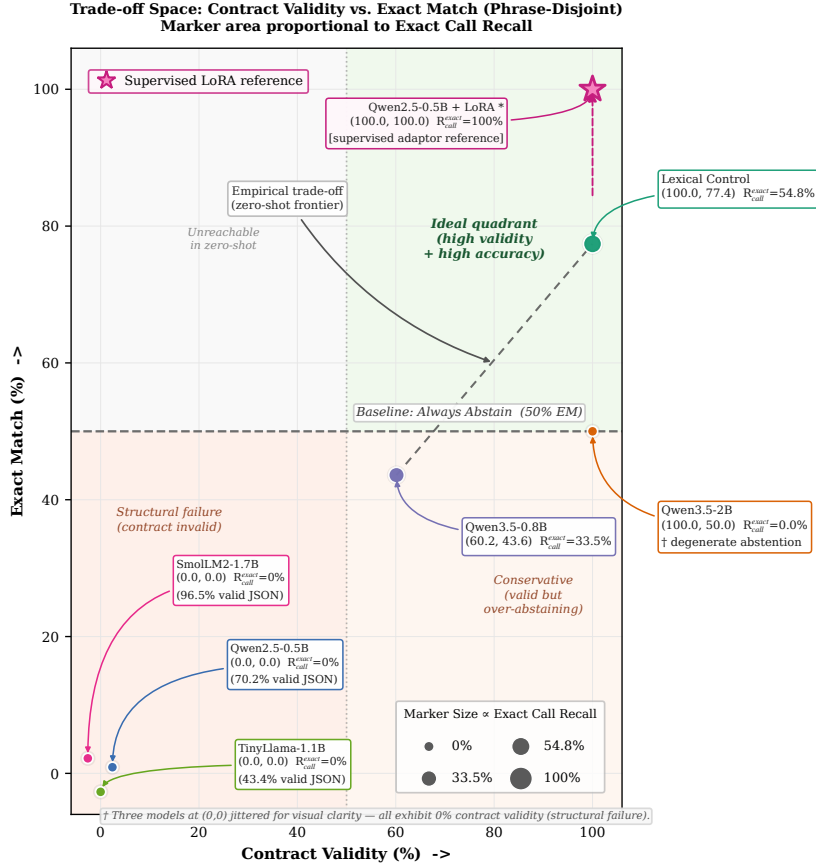
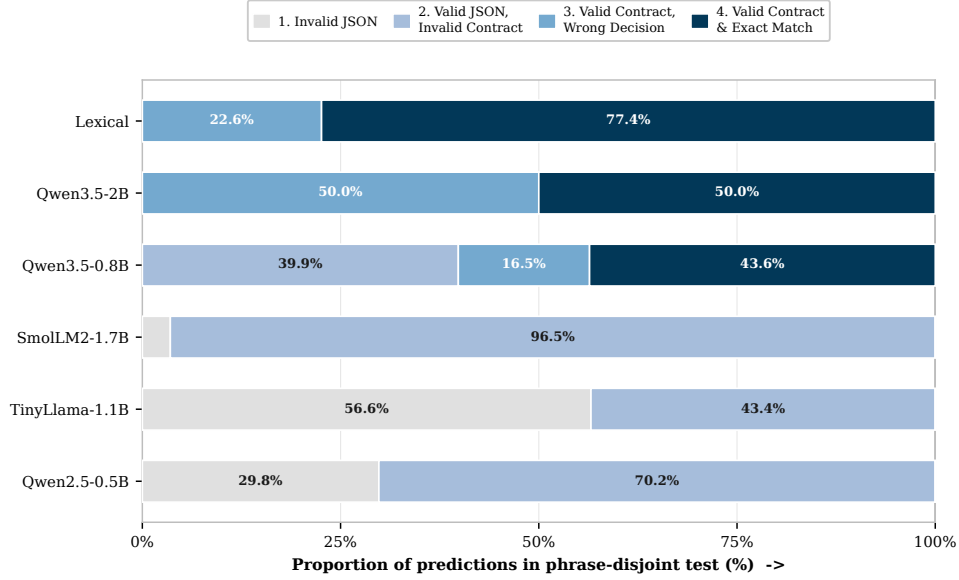


Figure 1. Trade-off space on phrase-disjoint test. Marker area $\propto$ exact call recall. The lexical baseline dominates zero-shot SLMs; Qwen3.5-2B collapses to 100% abstentions.

Figure 2 reveals the exact point of structural failure: SmolLM2-1.7B produces 96.47% syntactically parseable JSON, but 0% valid contract objects due to schema deviations (missing action/tool keys or invalid parameters). Unconstrained zero-shot decoding is insufficient for sub-3B models to adhere to rigid task schemas.



Stages: 1 Invalid JSON -> 2 Valid JSON but invalid contract -> 3 Valid contract but wrong decision -> 4 Valid contract and exact match. Values shown for slices $\geq 8\%$.

Figure 2. Structural cascade decomposing outputs into: (1) invalid JSON, (2) valid JSON but invalid schema, (3) valid contract but wrong call, and (4) exact match.

3. Discussion, Adaptation, and Conclusions

Two orthogonal techniques address this bottleneck: (i) **Grammar-Constrained Decoding** (e.g., GBNF in `llama.cpp`, Outlines [9]), which forces syntactic compliance by construction (collapsing stages 1–2 of Figure 2 to zero, though semantic discrimination still relies on the model); and (ii) **Supervised LoRA Adaptation** [8]: an in-distribution Qwen2.5-0.5B LoRA adaptor reaches 100% contract validity and 100% exact match, demonstrating that the contract is readily learnable with parameter-efficient fine-tuning.

In conclusion, zero-shot structured routing without grammar guidance is fragile in the ultra-small regime. Benchmarks must decouple syntactic validity from semantic utility, audit template leakage, and incorporate transparent non-neural baselines. Extending this evaluation to multi-domain SLU benchmarks like SLURP [2] and compositional parsing corpora like STOP [3] represents an essential next step.

References

- [1] L. Lugosch et al. Speech model pre-training for spoken language understanding. *Interspeech*, 2019.
- [2] E. Bastianelli et al. SLURP: A Spoken Language Understanding Resource Package. *EMNLP*, 2020.
- [3] P. Tomasello et al. STOP: A dataset for spoken task-oriented semantic parsing. *IEEE SLT*, 2023.
- [4] Z. Liu et al. MobileLLM: Optimizing sub-billion parameter models for on-device. *ICML*, 2024.
- [5] P. Zhang et al. TinyLlama: An open-source small language model. *arXiv:2401.02385*, 2024.
- [6] L. Lozhkov et al. SmolLM2: When Smol Goes Big. *arXiv:2502.02737*, 2025.
- [7] Qwen Team. Qwen3.5 model card. *Hugging Face*, 2026.
- [8] E. J. Hu et al. LoRA: Low-rank adaptation of large language models. *ICLR*, 2022.
- [9] B. T. Willard and R. Louf. Efficient guided generation for large language models. *arXiv:2307.09702*, 2023.